



Review article

Prospective validation of deep-learning algorithms for diabetic retinopathy screening: A systematic review and meta-analysis

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ABSTRACT

Deep-learning (DL) algorithms are widely promoted for diabetic-retinopathy (DR) screening, yet their prospective diagnostic accuracy is not well defined. PubMed, EMBASE and ClinicalTrials.gov were searched to April, 2025, for prospective evaluations of DL systems using color-fundus images. Two reviewers screened records, extracted data, and applied QUADAS-2. Hierarchical bivariate random-effects models produced pooled sensitivity and specificity for referable and vision-threatening DR, analyzed separately at patient and eye level. Twenty-one prespecified moderators were explored with uni- and multi-variate meta-regression; publication bias was assessed with Deeks' test. Seventy-three studies from 23 countries (255,330 examinations) met the criteria. Pooled patient-level sensitivity was 0.94 (95 % CI 0.92–0.95) and specificity 0.90 (95 % CI 0.87–0.93); eye-level values were 0.93 (95 % CI 0.91–0.95) and 0.94 (95 % CI 0.92–0.96). DR subtype, retinal-field strategy, camera form factor, and prevalence independently explained heterogeneity ($p < 0.05$). Performance matched or exceeded pivotal FDA trials (IDx-DR, EyeArt). AI gradability was ≥ 95 % in 60 % of cohorts, including handheld and smartphone systems. DL-based DR screening achieves consistent, high accuracy across devices and care settings, enabling scalable deployment in primary care, pharmacies, and mobile clinics. Quality assurance and ongoing monitoring are essential to maximize population-level benefits.

1. Introduction

Diabetic retinopathy (DR) is the leading cause of preventable blindness among working-age adults globally, representing a major public health concern that is intensifying alongside the worldwide diabetes epidemic. The International Diabetes Federation estimates that approximately 463 million people were living with diabetes in 2019—a number projected to rise to more than 700 million by 2045.⁹⁹ Among these, 1 in 3 individuals will develop DR, and a substantial proportion will progress to vision-threatening diabetic retinopathy (VTDR) or diabetic macular edema (DME) if not detected and treated in time.^{7,104} Early identification of referable diabetic retinopathy (rDR) is crucial to prevent irreversible visual impairment and reduce long-term societal and economic burdens.^{27,86} Although clinical guidelines in many

countries recommend annual dilated fundus examinations for patients with diabetes, adherence to screening remains disconcertingly low in real-world practice—often below 50 %.^{19,95} Missed or delayed diagnoses contribute directly to increased incidence of advanced retinopathy, late referrals, and avoidable vision loss. The screening gap is especially pronounced in underserved regions, where access to ophthalmic care is limited by geography, infrastructure, and workforce shortages.⁷⁷ These persistent barriers underscore the need for innovative and scalable screening strategies capable of reaching large populations efficiently and equitably.

Over the past decade, artificial intelligence (AI)—particularly deep learning (DL) based on convolutional neural networks (CNNs)—has emerged as a transformative force in medical image analysis. In DR screening, DL algorithms now outperform classical computer-vision

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